Session 4.
Random variables and distribution functions.

Pre course: Statistics.

Pablo Herrera G.



Hochschule
Bonn-Rhein-Sieg
University of Applied Sciences

Agenda

- Distribution viz – histograms (again).
- Distribution viz – density plots.
- Shape of a distribution
- Key features within a distribution.
- General description.
- Applied summary.
- Exercise on density

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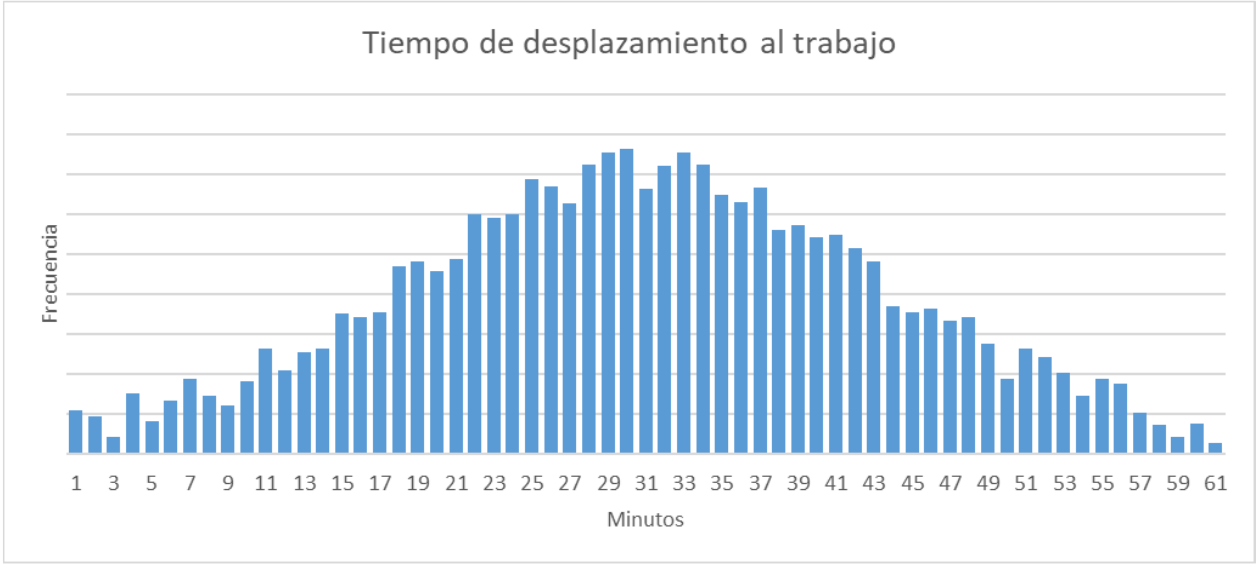
Histograms (again).

- X axis: continuous variable in ranges or bins (equal width intervals) used to group data.
  - Bins with equal width intervals
- Y axis: can be number of people but also proportions.



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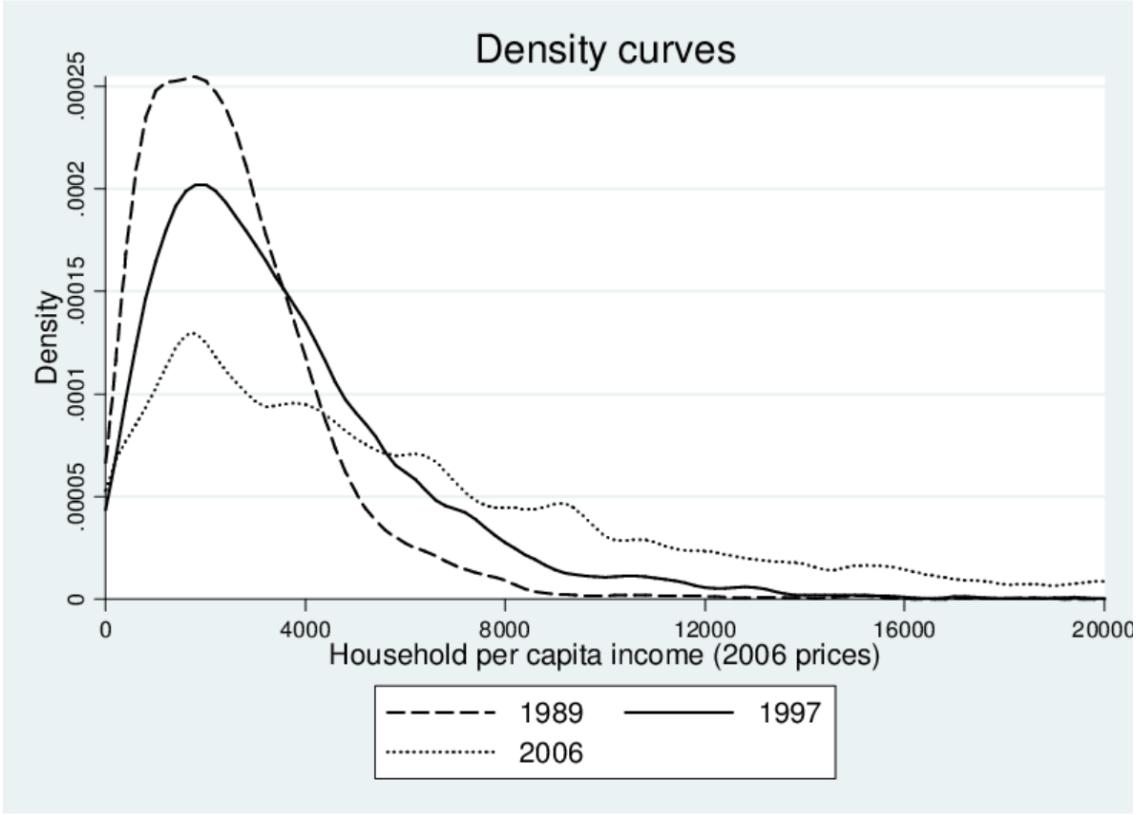


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Density plots.

- Smoothed curve that shows the distribution of a quantitative variable.
- X-axis shows the variable's values.
- Y-axis shows the density, not counts.
- Source: Source: Bonnefond, C., & Clément, M. (2012). An analysis of income polarisation in rural and urban China. Post-Communist Economies, 24(1), 15–37. https://doi.org/10.1080/14631377.2012.647972

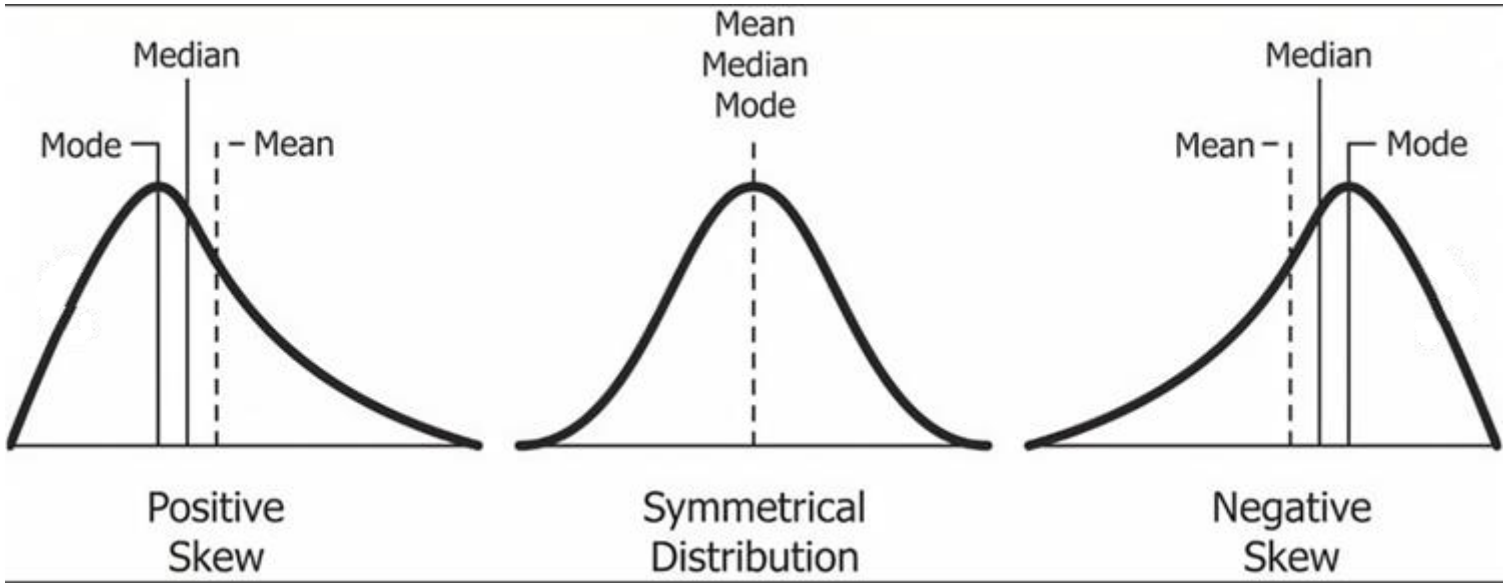


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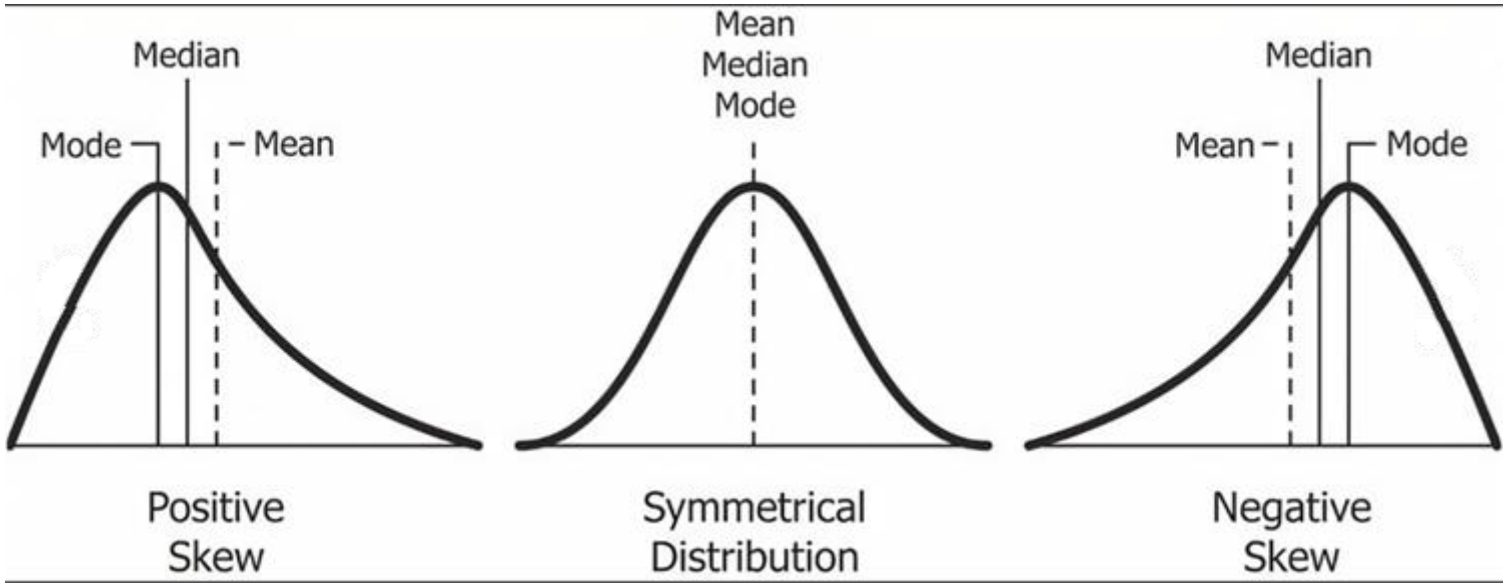
Shape of a distribution.

- Symmetry vs. asymmetry (central tendency measures).
- Skewness (right, left).
- Other shapes: bimodal, uniform.



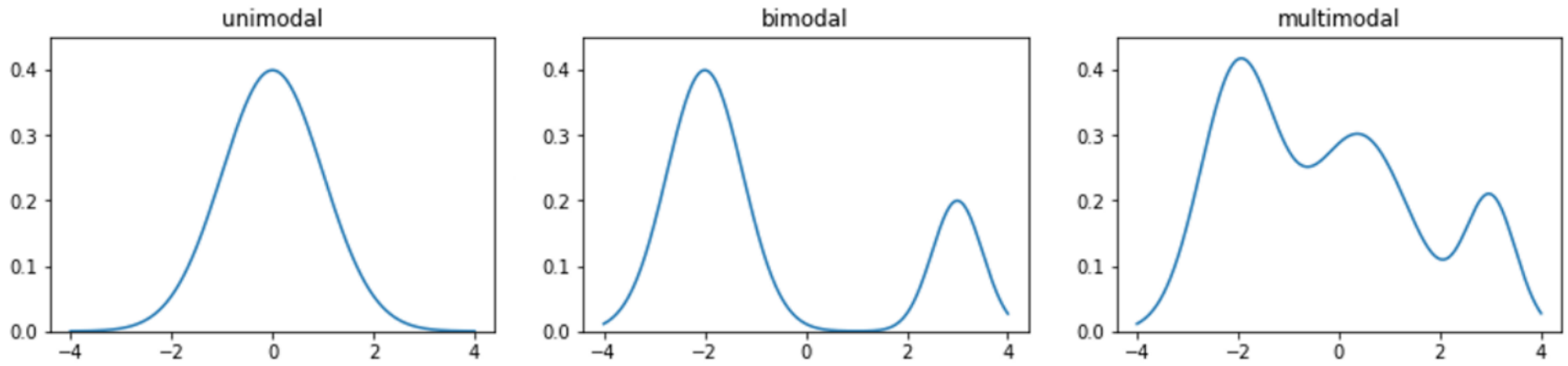
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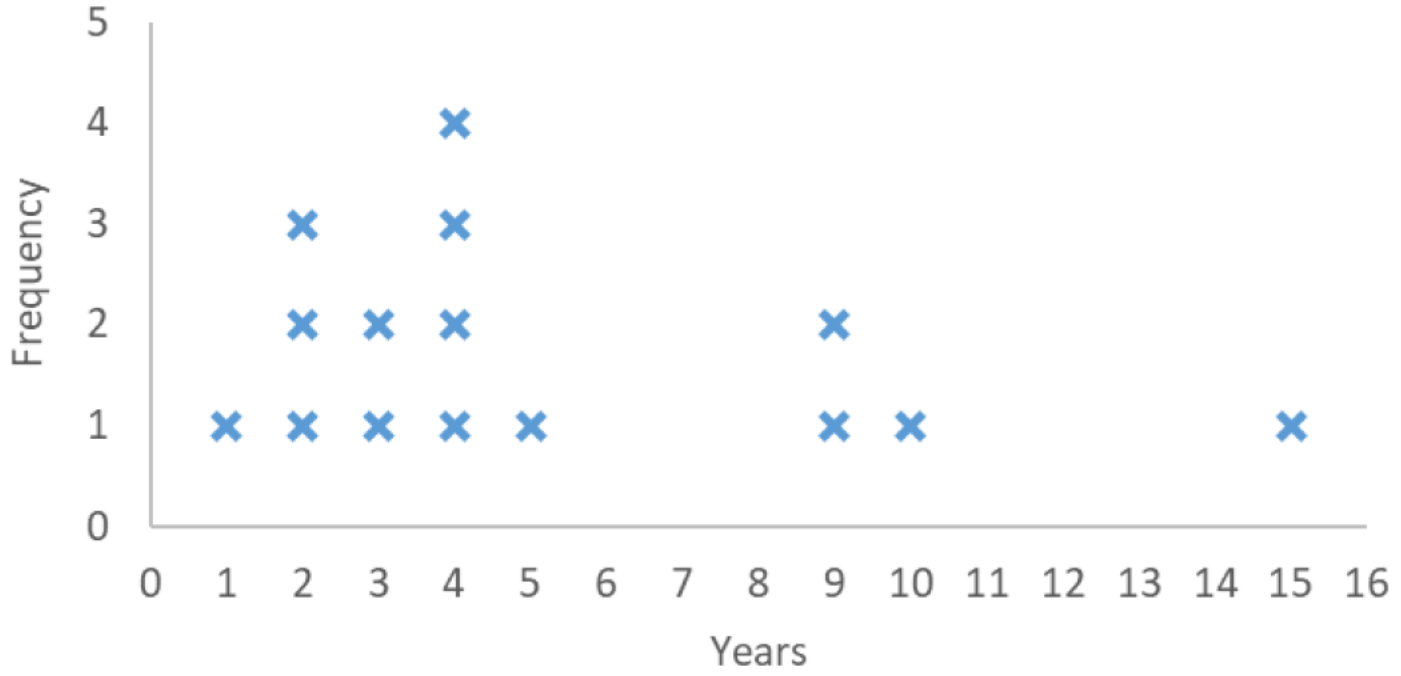


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Key features within a distribution.

- Peaks (modes).
- Clusters.
- Gaps.
- Outliers.



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General description.

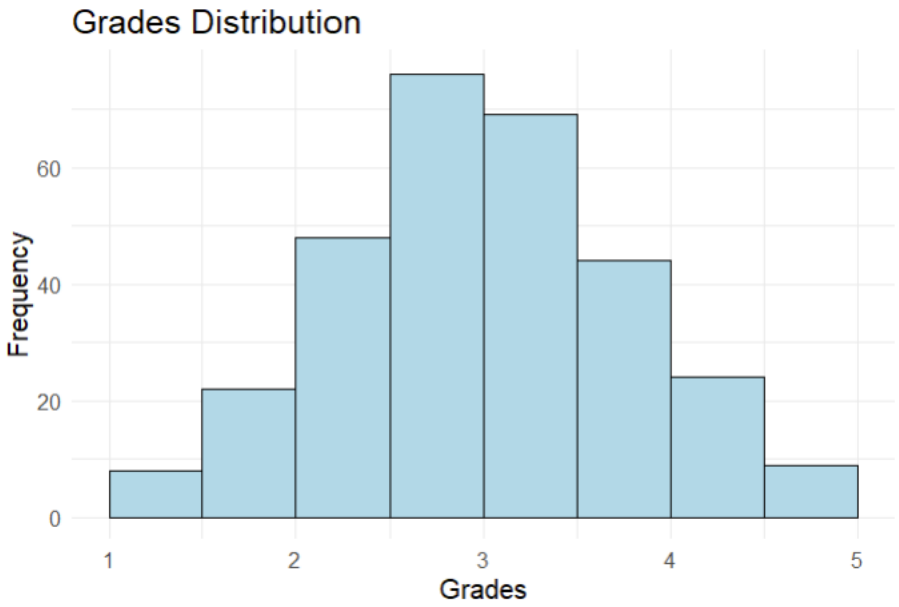
- Shape (skewed left, symmetric, etc.).
- Center (mean, median).
- Spread (range and other dispersion measures).
- Outliers (yes/no, how identified).
- Other components (peaks or modes, clusters, gaps).

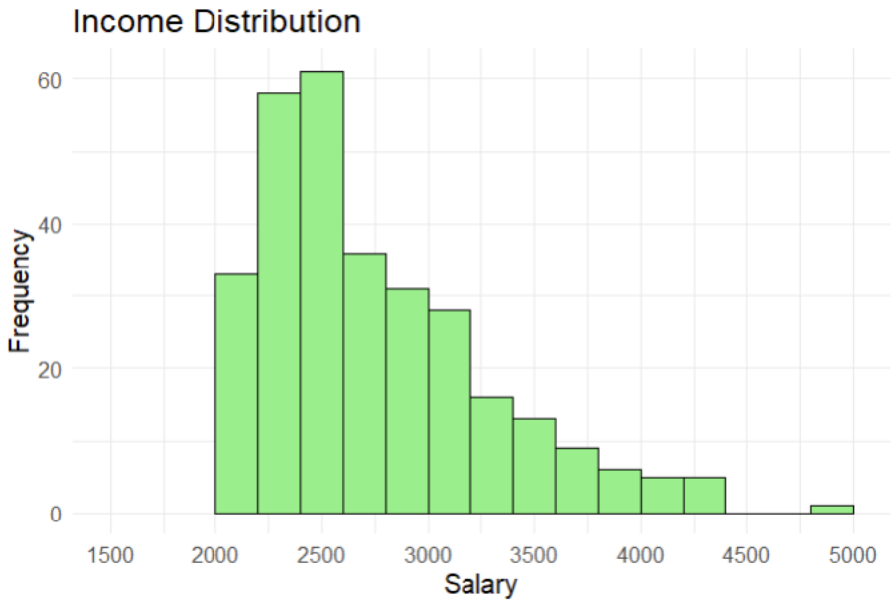
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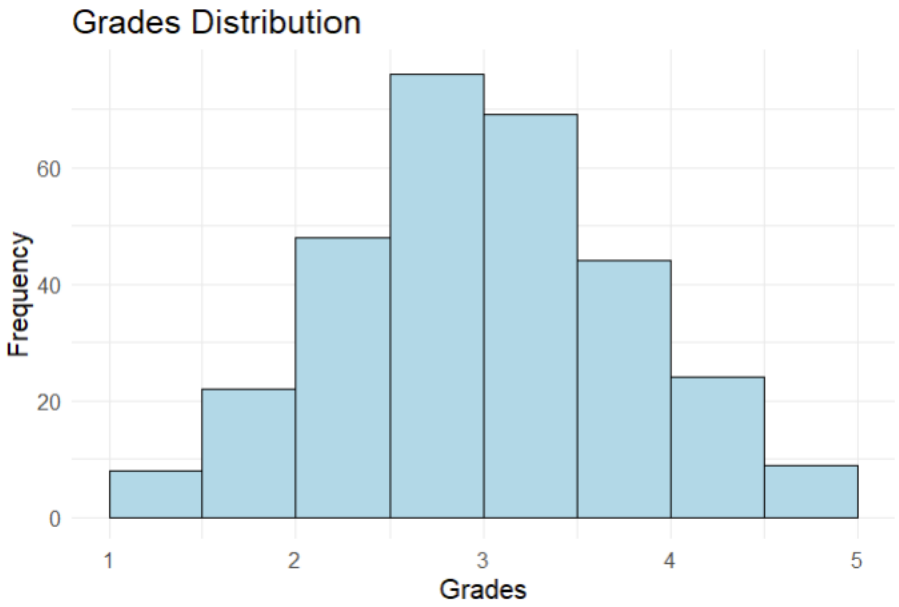
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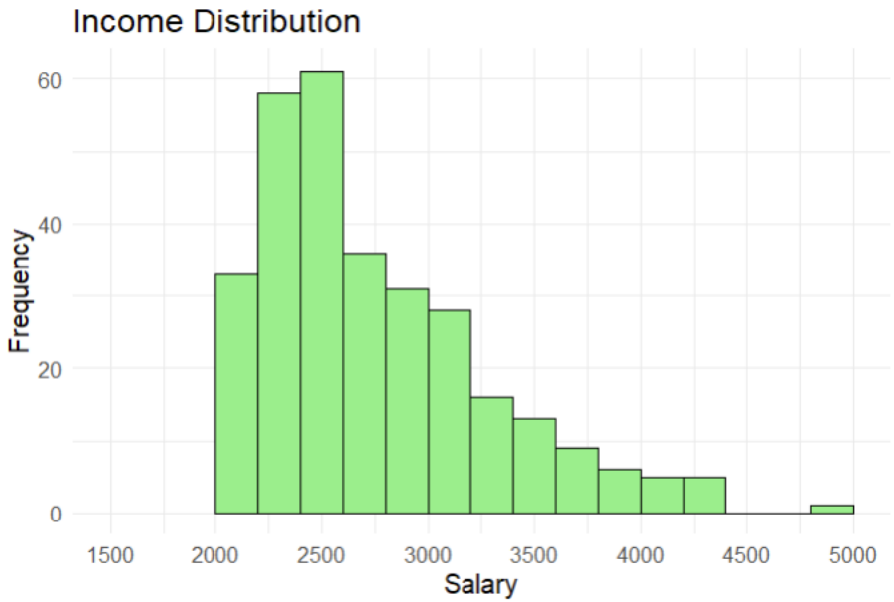




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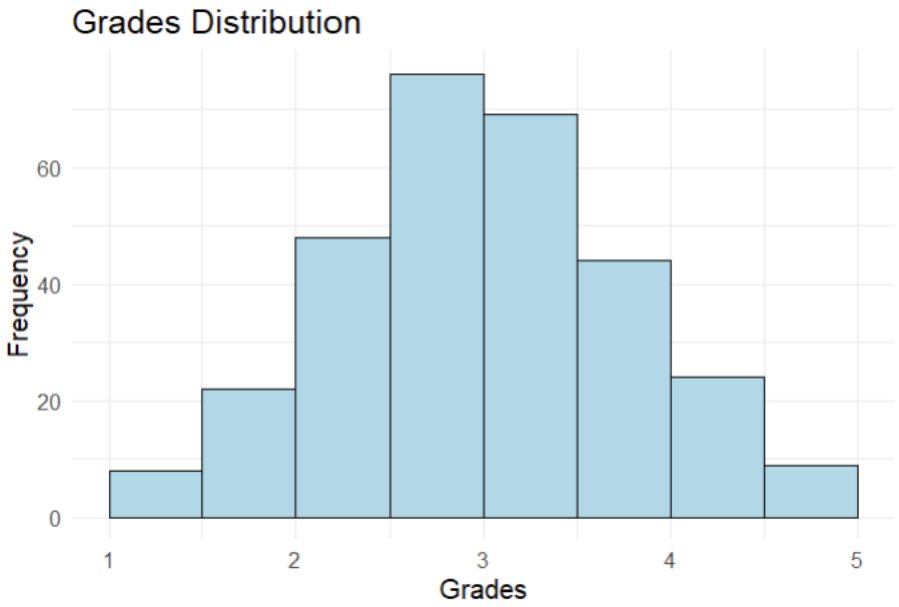
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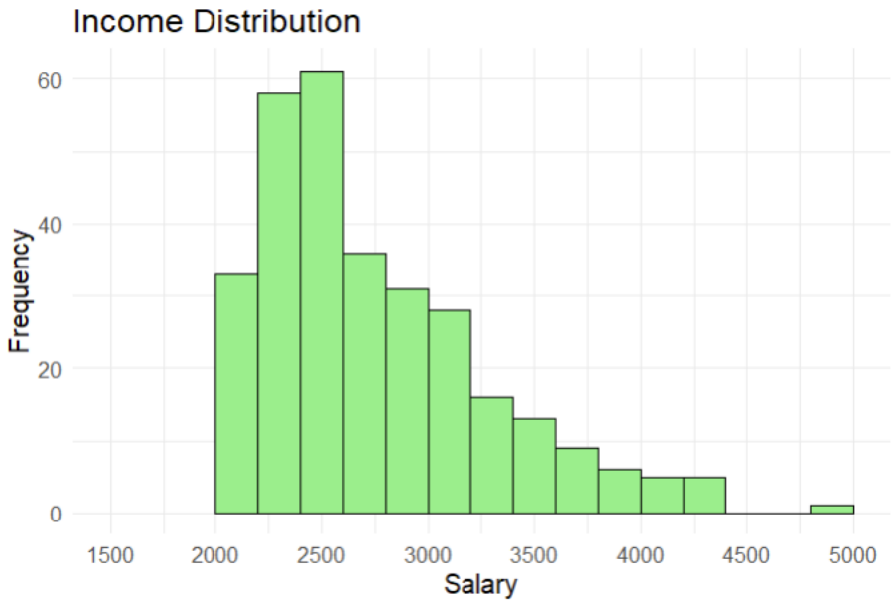




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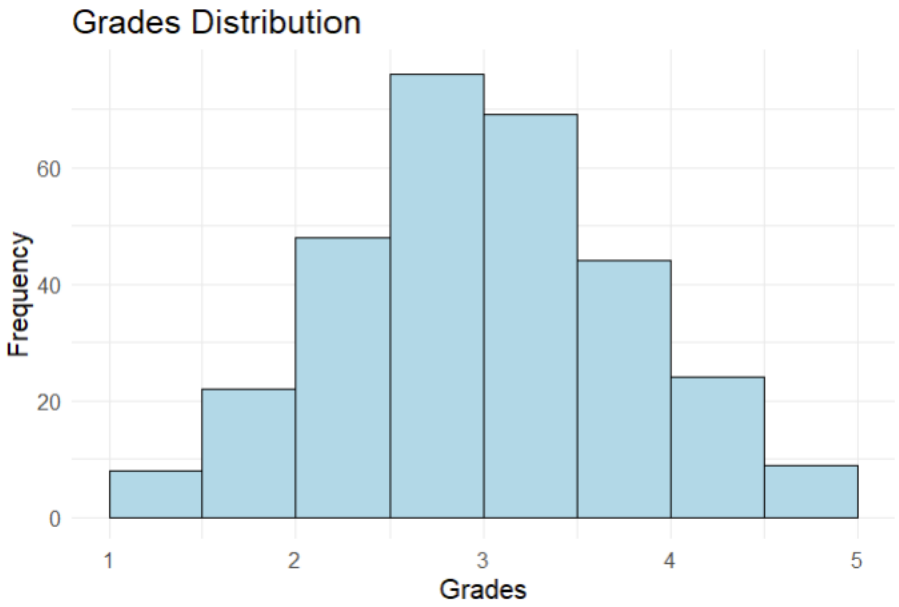
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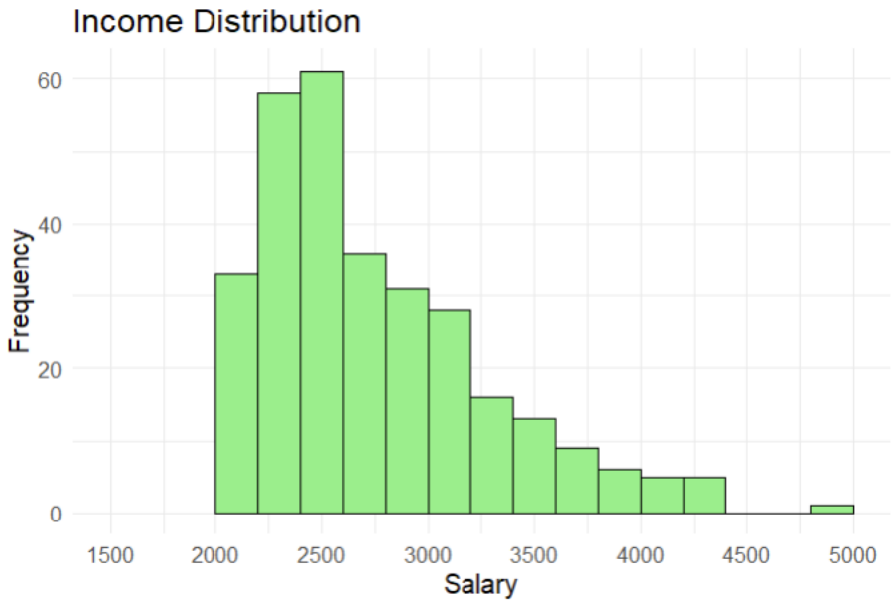




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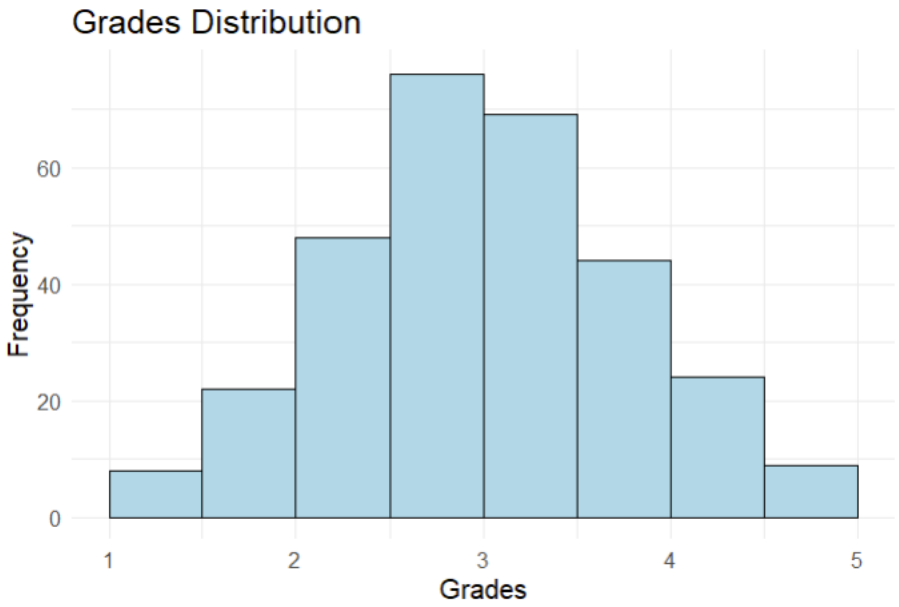
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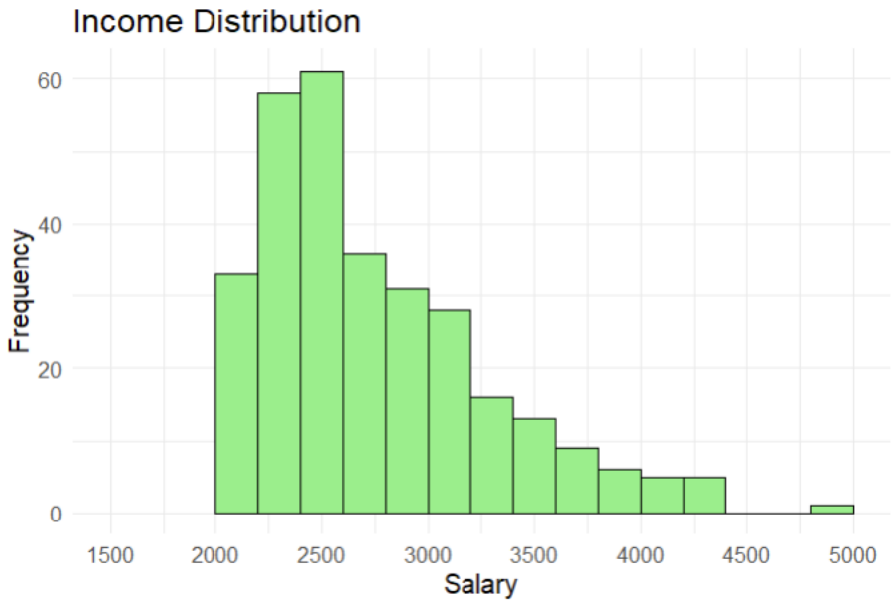




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Exercise 3.

a) In which period was more share of the population with an income per capita of 4000 or less?

b) Which period shows more equal incomes? Explain why.

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